

Final examination on Compilers

Christian Rinderknecht

13 December 2005

Question. Consider the following Lex specification:

```
%{
#include<string.h>
char* keyword[] = {"else", "if", "return", "then"};
%}
num  (\.[0-9]+|[\+-]?[0-9]+(\.[0-9]+)?)(E[\+-]?[0-9]+)?
id   [A-Za-z]([_]*[A-Za-z0-9])*
ws   [ \n\t]+
%%
{num} { printf ("num<%s>\n",yytext); }
{id}  { int index = 0;
        while (index <= 3
                && strcmp(keyword[index],yytext)) index++;
        if (index == 4) printf("id<%s>\n",yytext);
        else printf("kwd<%s>\n",yytext);
      }
{ws}  {}
"+"   { printf ("plus<>\n"); }
%%
```

Some possible transition diagrams corresponding to the regular expressions `num`, `id` and `ws` are respectively given in figure 1, 2 and 3.

Define the meaning of the pointers $\downarrow$, $\uparrow$ and $\uparrow\uparrow$ presented in class and show how the input

`return__x+_ .5E2+y_0`

where `_` represents a blank character, is analysed using the transition diagrams. Make clear of which diagram you are referring to.

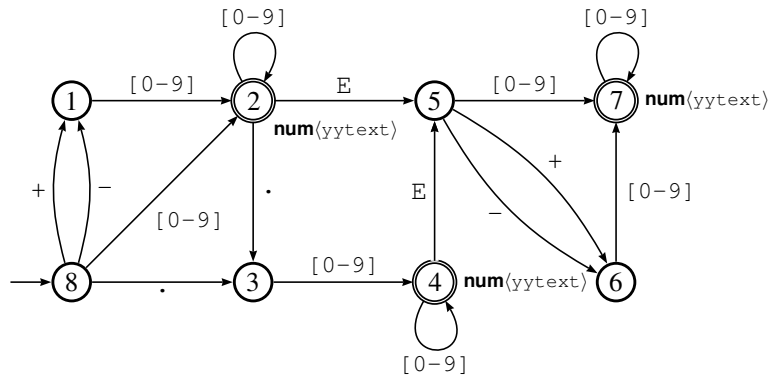


Figure 1: Transition diagram for regular expression `num`

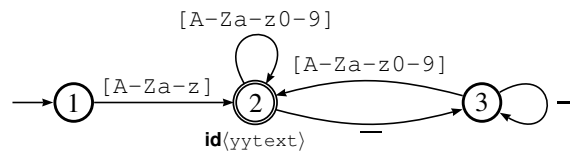


Figure 2: Transition diagram for regular expression `id`

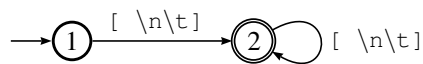


Figure 3: Transition diagram for regular expression `ws`